

Aiyana Brown

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SUMMARY

Data Science Master's student who approaches data like a consultant: gathering the real question behind a request, then turning messy data into clear, actionable recommendations. Comfortable with the full stack, from backend data engineering (SQL, pipelines, database architecture) to frontend storytelling (dashboards, visualization, stakeholder communication). Motivated by people-focused work to improve my community.

SKILLS

Software: SQL (TSQL, Postgres), Power BI, Python (scikit-learn, pandas), R (tidyverse), dbt, Airflow, Grafana, Excel, GitHub, Jira

Technical: Data visualization & dashboards, data storytelling, data engineering, machine learning (regression, classification), statistical modeling, cost-benefit analysis

Professional: Stakeholder collaboration, project management, communication, organization, written and verbal reporting

EDUCATION

M.S. Data Science (3+1 Program), Willamette University | Salem, OR

Expected Sept 2026

B.S. Data Science, Economics Minor, Willamette University | Salem, OR | 4.0 GPA

EXPERIENCE

Data Analysis & Strategy Intern, Salem-Keizer Public Schools | Salem, OR

Jun 2026 – Present

- Identified a 10-point gap between the district's 2030 educational success target and its actual trajectory, using a 5-year effect size projection (Cohen's h) that is now driving a push to revise strategic goals with the state
- Served as an internal data consultant across departments, reconciling inconsistent SQL Server/Power BI metrics and advising the finance team on visualization design to more clearly communicate investment portfolio forecasts to stakeholders
- Directly shaped a new teacher-facing data platform's rollout toward better student outcomes by analyzing usage patterns and thematically coding qualitative survey feedback in R, producing reports that identified key features to add and data inconsistencies to fix
- Modernized how the district stores and trusts its data by helping build a 4-database vault architecture and onboarding new data sources through SQL-based ETL processes, replacing an old system to ensure integrity of historical records and connection speed

Statistical Engineering Researcher, Willamette University | Salem, OR

Jun 2025 – Dec 2025

- Reduced inspection waste by building interactive dashboards to compare cost trade-offs and sampling outcomes, consulting stakeholders on optimal inspection strategies
- Improved quality-control and cost-optimization processes by developing an open-source Six Sigma package for sampling audits and financial analysis
- Enabled data-informed decisions for engineers and managers by collaborating with faculty and peers to translate analytical findings into clear business recommendations

Data Science Q.U.A.D. Tutor, Willamette University | Salem, OR

2024 – Dec 2025

- Supported students' learning across the data lifecycle (cleaning, modeling, visualization) in R and Python; strengthened analytical reasoning and ability to connect quantitative findings to real-world context, emphasizing clarity and ethical use of data

Library Student Manager, Willamette University | Salem, OR

Sept 2023 – Jun 2026

- Managed operations for 100+ patrons per shift while coordinating staff and resolving issues; used data to improve resource accessibility by analyzing student feedback and usage trends

PROJECTS

Exploration of Income's Effect on Education Quality

GIS, R, Spatial Analysis | Fall 2024

- Analyzed relationships between income and education quality using regression and spatial analysis, identifying key socioeconomic drivers of educational success in Oregon
- Visualized inequities using ArcGIS and R, presenting findings through interactive maps and reports to inform policy interventions in underserved areas

Healthcare Spending Efficiency Analysis

Python, R, Regression | Fall 2025

- Evaluated healthcare spending efficiency through analysis of 30+ years of regional data from multiple sources
- Identified effective policy and healthcare investment strategies for specific populations and geographic locations through regression modeling and segmentation
- Communicated findings through reports and visualizations, including limitations and recommendations

Telecom Customer Churn Prediction Project

Python, Scikit-learn, XGBoost | Spring 2026

- Developed machine learning models to predict churn, including preprocessing, feature engineering, and model training
- Evaluated performance (ROC AUC, precision/recall) and compared algorithms, achieving up to 95% accuracy and identifying key drivers of retention